A Systematic Literature Review of Large Language Models and Small Language Models in Hybrid, Multi-Agent, and Monolithic Architectures

Doğukan Topçu[1], Abdulbaki Çeltik[1], Berkhan Kargılı[1], Meriç Kelemeden[1], Onur Demirörs[2], and Hüseyin Ünlü[2]

[1] Izmir Institute of Technology, Izmir, Turkey
dogukantopcu@std.iyte.edu.tr
abdulbakiceltik@std.iyte.edu.tr
berkhankargili@std.iyte.edu.tr
merickelemeden@std.iyte.edu.tr
[2] Izmir Institute of Technology, Izmir, Turkey
Advisors
onurdemirors@iyte.edu.tr
huseyinunlu@iyte.edu.tr

Abstract. Large language models (LLMs) now define the state of the art in many reasoning, generation, and decision-support tasks, but their deployment cost, energy demand, latency, and hardware requirements remain substantial. Small language models (SLMs) provide an attractive alternative for edge deployment, privacy-sensitive settings, and task-specific optimisation, yet their capabilities are uneven across domains. This paper presents a systematic literature review of 54 studies on LLMs, SLMs, and SLM–LLM collaboration, with emphasis on hybrid, multi-agent, retrieval-centered, and monolithic architectures. The review is guided by four research questions addressing comparative performance, computational efficiency, orchestration mechanisms, and domain suitability. Using a PRISMA 2020-guided review design and a Web of Science-centered search process, we synthesize evidence on model choices, architectural patterns, datasets, metrics, baselines, latency, cost, energy, and reported limitations. The reanalysis of the corpus shows a strong shift toward hybrid and multi-agent designs together with a clear reporting asymmetry in which accuracy is common but cost and energy are rarely quantified. The findings indicate that SLMs are most competitive when they are paired with routing, retrieval, planning, or fusion mechanisms and when the problem space is narrow, structured, or domain-specific.

Keywords: systematic literature review · large language models · small language models · hybrid architectures · multi-agent systems · edge-cloud inference

1 Introduction

Large language models (LLMs) excel at reasoning, coding, and multimodal tasks [52, 47, 46]. However, due to the high hardware requirements (RAM, GPU, etc.) they demand, these models are usually run only on cloud-based infrastructures or very advanced local systems [39, 51]. This creates a significant bottleneck for system developers seeking to reduce cloud dependency, ensure data privacy, and avoid high operational costs [1, 8].

As a result of these limitations, small language models (SLMs) are emerging as a strong alternative. Their ability to run on local machines provides a significant advantage in terms of data security and deployment autonomy, while also offering substantially lower operational costs than cloud-scale LLM setups [1, 8]. Moreover, recent studies show that SLMs can be adapted to specific domains via fine-tuning or distillation, enabling competitive performance in specialized tasks [3, 60, 37]. In practical systems, SLMs are also attractive because they integrate well with explicit control modules (e.g., rule-based filters, routing/verification thresholds, or planners), improving controllability and reliability beyond prompt-only solutions [8, 51, 10]. However, SLMs still struggle to match the broad expertise and robust reasoning of massive frontier models, especially on tasks requiring deep contextual understanding and extensive world knowledge [52, 12]. Therefore, framing the problem as a simple "SLM or LLM?" competition is no longer sufficient; instead, current research investigates when a single SLM is adequate, when an LLM is necessary, and when hybrid or agent-based architectures are preferable [2, 16, 32].

The existing literature provides concrete, domain-specific examples showing that the choice between SLMs, LLMs, and hybrid designs is primarily a systems decision. In telecommunications, Symbiotic Agents reports up to a fivefold reduction in decision error and near-real-time operation (82 ms loops) with up to 99.9% lower GPU overhead using SLM variants [20]. In medicine, self-reflective pipelines pair a compact generator with an LLM-based judge to iteratively refine outputs and improve diagnostic reliability [21].

These developments provide practical alternatives for scenarios where relying on a single monolithic LLM is either insufficient for task demands or impractical under strict resource constraints. Accordingly, recent work increasingly targets measurable improvements beyond raw accuracy, including reduced computational load, lower operational cost, and tighter latency budgets [2, 51, 20, 39]. Across domains, researchers have reported which hybrid design patterns are effective for specific task classes—for example, selective edge–cloud verification, expert routing, and retrieval- or planner-centered pipelines—and how these choices translate to real-world deployments [2, 6, 4, 10].

Motivated by these trends, this review analyzes when a single SLM is sufficient, when an LLM remains necessary, and when hybrid or multi-agent architectures provide the best trade-off between capability and efficiency [1, 3, 2, 16]. We further examine domain conditions under which fine-tuned SLMs can realistically replace or complement larger models, and where they still fall short, synthesizing recurring architectural motifs across telecom, medical, and cybersecurity settings [20–22].

The main contributions of this review are as follows: (1) It provides a synthesized overview of 56 studies covering cloud-based monolithic LLMs, hybrid SLM–LLM architectures, the performance of fine-tuned SLMs versus large models, and multi-agent workflows. (2) It reframes the literature around four distinct practical research questions: Performance, Efficiency, Architecture, and Domain Suitability. (3) It creates an analytical review matrix that reveals the commonalities and differences among the studies, mapping which specific research question each study addresses.

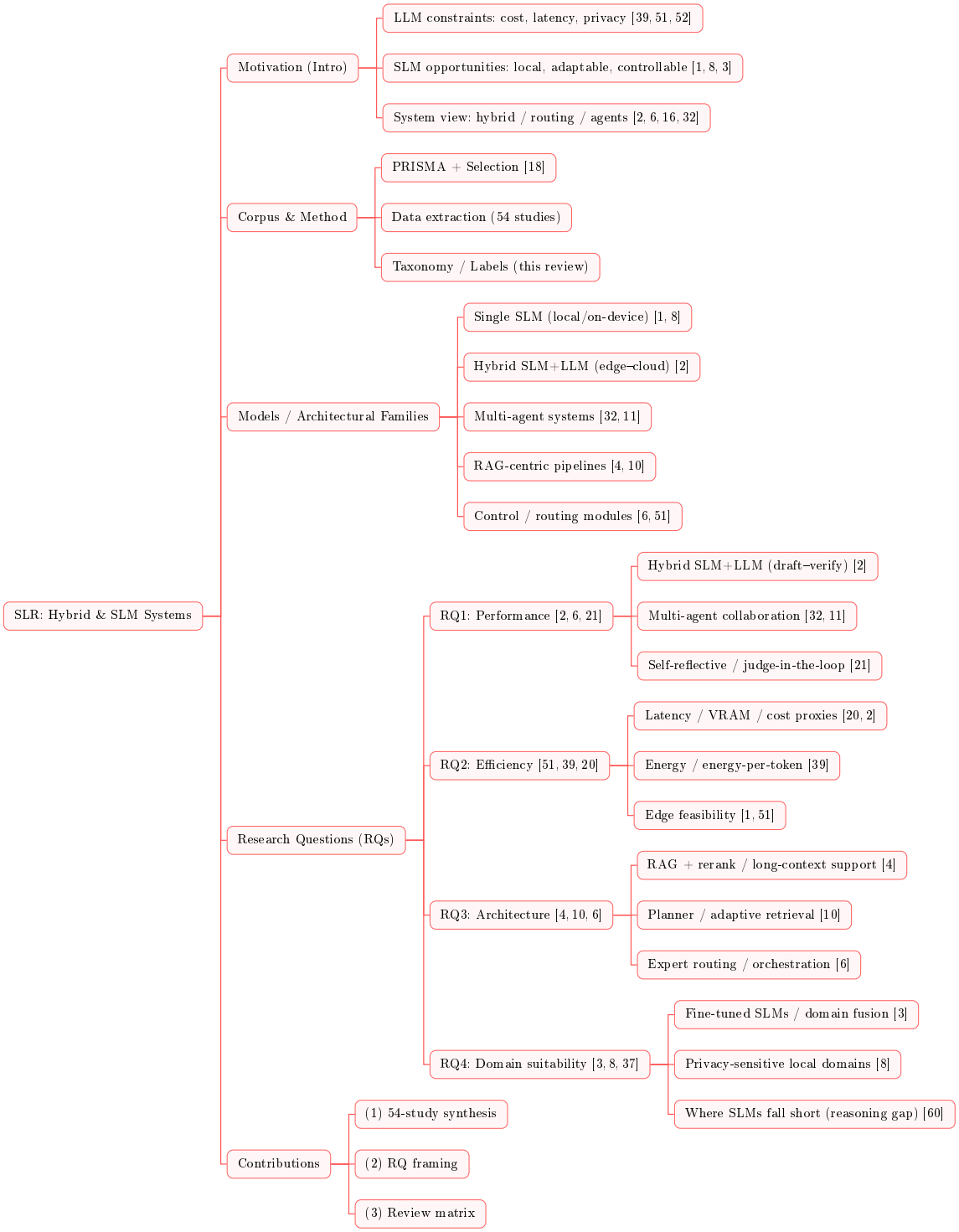

Fig. 1. Overview of the review structure and taxonomy used in this SLR (synthesized from the reviewed corpus; reporting guided by PRISMA [18]).

2 Related Works

Prior review literature on Small Language Models (SLMs) and SLM–LLM collaboration can be grouped into four highly relevant works that overlap with our scope at different levels of methodology and emphasis. The closest methodological competitor is *State of the Art and Future Directions of Small Language Models: A Systematic Review*, a PRISMA 2020-guided SLR that sources studies from multiple databases and provides a reproducible selection protocol [56]. While strong in transparency, it is broader in topic focus (SLMs in general) and its coverage window ends around early 2025, limiting visibility into the 2025–2026 wave of hybrid and agent-based systems.

The remaining three works are narrative surveys with substantial thematic overlap. *A Comprehensive Survey of Small Language Models in the Era of LLMs* offers wide coverage of SLM definitions, training techniques, applications, collaboration with LLMs, and trustworthiness, but it does not follow a systematic and reproducible selection protocol [57]. *A Survey on Collaborative Mechanisms Between Large and Small Language Models* provides a technically detailed taxonomy of collaboration patterns (e.g., routing, auxiliary modules, fusion, and distillation) with an emphasis on edge-device requirements, yet its analysis is primarily mechanism-centered rather than corpus-centered [58]. Finally, *A Survey on Collaborating Small and Large Language Models for Performance, Cost-effectiveness, Cloud-edge Privacy, and Trustworthiness* overlaps strongly with our themes (performance, cost, privacy, trustworthiness) but remains a narrative survey without a formal PRISMA-guided study selection and extraction schema [59].

Across these reviews, the gap is not a lack of attention to SLMs or collaboration, but a difference in analytical emphasis and evidential grounding. Existing surveys typically answer what kinds of models or mechanisms exist; they less often align evidence across domains and report comparable operational metrics. In contrast, our study is positioned as a PRISMA-guided, engineering-oriented synthesis that (i) treats domain suitability as a first-class axis, (ii) maps research questions to extraction fields so claims remain traceable, and (iii) emphasizes quantitative indicators (accuracy, latency, cost, and energy) to support deployment-relevant decisions [18].

3 Research Methodology

3.1 Information Sources and Search Strategy

To capture a comprehensive and high-quality corpus of contemporary AI architectures, the literature search was conducted utilizing the Web of Science (WoS) Core Collection. Utilizing WoS as the primary search engine allowed for the retrieval of indexed peer-reviewed journal articles, conference proceedings, and highly cited preprints within a single dataset. The search timeframe was restricted to publications between 2023 and 2026 to reflect the rapid, recent emergence of small and agentic language models.

The search strategy was designed to capture the intersection of model size, architectural design, and system evaluation. The following Boolean query was applied to the title, abstract, and keyword fields:

> ("Large Language Model*" OR "LLM" OR "Generative AI" OR "Foundation Model*") AND ("Small Language Model*" OR "SLM" OR "Compact Model*" OR "Lightweight Model*" OR "TinyLlama" OR "Phi-3" OR "Mistral" OR "Qwen") AND ("Multi-agent" OR "Hybrid" OR "Orchestration" OR "Routing" OR "Ensemble" OR "Agentic") AND ("Performance" OR "Accuracy" OR "Efficiency" OR "Latency" OR "Cost" OR "Benchmark")

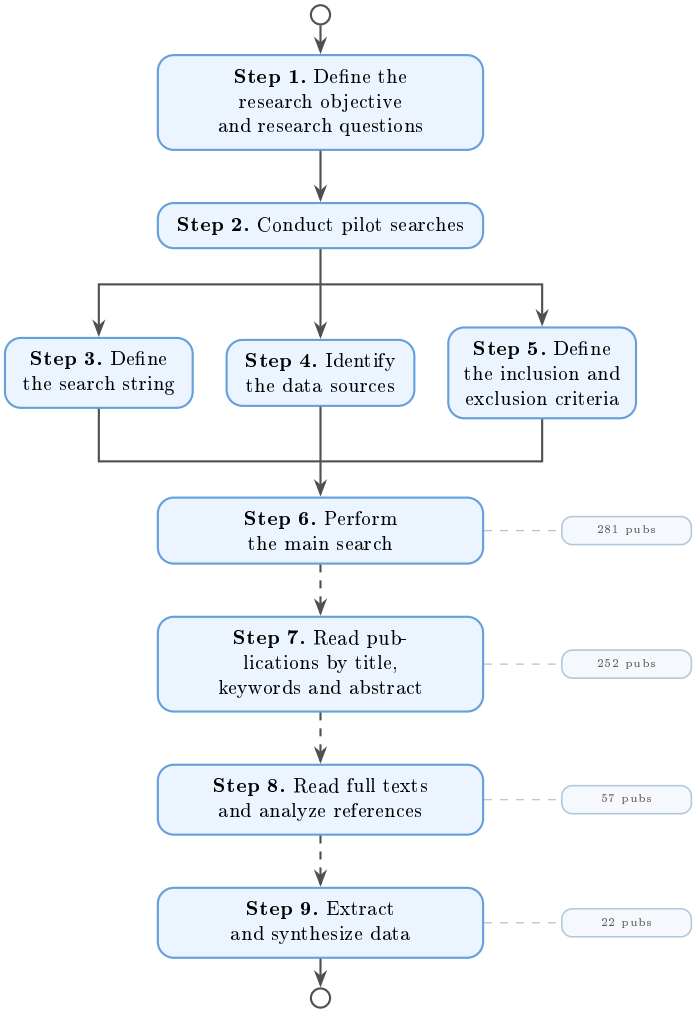

Fig. 2. Systematic literature review process

3.2 Eligibility Criteria

To systematically isolate systems-level engineering research from the broader generative AI literature, retrieved records were evaluated against strict eligibility criteria. A comprehensive breakdown of the inclusion and exclusion parameters applied during the screening phases is presented in Table 1.

Table 1. Inclusion and Exclusion Criteria for Study Selection

Criteria	Inclusion (Include if...)	Exclusion (Exclude if...)
Architecture Scope	Studies evaluating SLMs alone or SLM and LLM collaborative architectures (e.g., Hybrid or Multi-agent).	Studies strictly comparing massive monolithic LLMs (e.g., GPT-4 vs. Gemini) without an SLM component.
Model Size	Architectures involving "small" models that strictly adhere to a parameter count of ≤ 7 Billion.	"Closed-box" or API-bound models where parameter count or architectural details are undisclosed (violates reproducibility).
Topic Focus	Focuses on quantitative system-level metrics such as performance, inference latency, memory, cost, or orchestration efficiency.	Focuses solely on ethical, sociological impacts, or legal regulations without technical benchmarks.
Data Quality	Contains empirical results and experimental validation (e.g., standard benchmarks or custom datasets).	Purely theoretical papers or non-empirical narrative surveys.
Timeframe	Published between 2023 and 2026 (capturing the recent emergence of SLM/Agentic architectures).	Published before 2023.

3.3 Study Selection Process

The study selection process adhered to the PRISMA 2020 guidelines [18]. The initial literature search yielded 116 records. Because the export was generated from a single, centralized database engine (Web of Science), zero duplicate records required manual removal. The progression of the screening process is illustrated in Figure 2.

The screening was conducted in two distinct phases to ensure methodological rigor. During the initial Title and Abstract screening, 22 records were excluded for lacking general relevance to the architectural scope. These exclusions primarily consisted of non-empirical narrative surveys, hardware scheduling studies lacking architectural orchestration, and studies operating in entirely separate domains (e.g., computer vision or 3D hand gesture recognition).

The remaining 94 records advanced to a rigorous Full-Text review. During this phase, an additional 40 studies were excluded after deeply evaluating their methodologies against the technical constraints. Exclusions at this stage occurred when the evaluated "small" models exceeded the strict

$\leq$ 7B parameter threshold, the evaluations relied on closed-box APIs that hindered reproducibility, or the methodologies failed to report quantitative system-level performance metrics.

To mitigate selection bias during both screening phases, a team consensus mechanism was utilized. Inclusion and exclusion decisions were made via majority vote among the reviewing authors. In the event of a tie, the conflict was resolved through collaborative discussion and escalation to the academic advisors. Following this two-stage screening process, a final corpus of 54 primary studies was established for data extraction and synthesis.

3.4 Data Extraction and Synthesis Strategy

To systematically address the research questions, a standardized data extraction protocol was applied to the final corpus of 54 primary studies. For each study, data was extracted into a structured matrix to capture both the functional design of the systems and the empirical evidence used to justify them.

The extracted fields included the specific SLM and LLM models utilized, the overarching architecture type (e.g., Hybrid, Multi-Agent, RAG, Ensemble), orchestration mechanisms, evaluation metrics, baseline models, and quantitative performance indicators (accuracy, latency, cost, and energy). To ensure analytical rigor, each extracted variable was explicitly mapped to one or more of the core Research Questions (RQs). This mapping is detailed in Table 2.

Table 2. RQ $\times$ data extraction matrix used in the review. A check mark indicates that the field directly supports the corresponding research question.

Extraction Field	RQ1	RQ2	RQ3	RQ4
Models Used (SLM)	✓			✓
Models Used (LLM)	✓			✓
Architecture	✓			✓
Orchestration Method			✓	✓
Agent Count		✓		✓
Task Type	✓		✓	
Dataset	✓		✓	
Metrics	✓		✓	✓
LLM Baseline	✓		✓	✓
SLM Baseline	✓		✓	✓
Accuracy	✓		✓	
Latency (ms)		✓	✓	✓
Cost ($)		✓	✓	✓
Energy		✓	✓	✓
Key Findings	✓	✓	✓	✓
Limitations		✓	✓	

Following data extraction, a narrative synthesis approach was utilized. Because the included studies operate across diverse domains (e.g., telecommunications, healthcare, software engineering) and report varying metrics, a purely statistical meta-analysis was not feasible. Instead, studies were grouped by their architectural taxonomy, allowing for a comparative analysis of how different orchestration methods and domain constraints impact the performance and efficiency trade-offs between large and small models.

4 Summary of the Studies

Before detailing the specific findings related to the research questions, this section provides a structural overview of the 54 primary studies included in this review. During the data extraction phase, 15 key characteristics were systematically captured for each paper to facilitate comparative analysis. Each extraction field is defined below:

- **Study ID:** A unique identifier (S1–S54) assigned to each paper for streamlined referencing throughout the text and appendix.
- **Title:** Identifies the name of the paper or proposed framework.
- **Citation/Ref:** The corresponding reference entry in the bibliography.
- **Year:** Indicates the year the study was formally published or released as a preprint.
- **SLM(s):** Lists the specific small language model(s) evaluated, fine-tuned, or deployed within the proposed system.
- **LLM(s):** Specifies the large language model(s) utilized either as a performance baseline, a teacher for distillation, or a cloud-based fallback.
- **Architecture (Arch.):** Describes the overarching structural design of the system, such as hybrid edge-cloud, multi-agent ensemble, or retrieval-augmented generation.
- **Orchestration Method:** Specifies the exact mechanism used to coordinate models and tasks, such as token-level routing, prompt-guided fusion, or majority voting.
- **Agent Count:** Indicates the number of distinct agents, roles, or expert models operating within the proposed system.
- **Task Type:** Details the specific downstream application, domain, or computational problem evaluated in the study.
- **Dataset:** Identifies the specific benchmarks, open-source datasets, or proprietary clinical/industrial data used to train and evaluate the system.
- **Metrics:** Outlines the primary evaluation criteria used to assess model performance, efficiency, and resource consumption.
- **LLM Baseline:** Reports the baseline performance score of the standalone large language model.
- **SLM Baseline:** Reports the baseline performance score of the standalone small language model.
- **Accuracy:** Presents the final performance or accuracy metric achieved by the proposed hybrid or SLM-centered architecture.
- **Latency (ms):** Captures the inference speed, time-to-first-token, or overall time delay reported by the authors.
- **Cost ($):** Highlights any reported financial savings regarding API usage or computational reductions relative to full LLM inference.
- **Energy:** Details power consumption, Joules-per-token, or carbon footprint implications, if measured by the authors.
- **Key Findings:** Summarizes the core takeaway and primary contribution of the study regarding SLM-LLM orchestration.
- **Limitations:** Highlights the constraints, failure modes, data dependencies, or hardware bottlenecks identified by the authors.

Because the full comparative analysis table is located in the online appendix, the mapping of Study IDs to their respective bibliographic citations is provided below for quick reference:

Primary Studies Mapping: S1: [1], S2: [2], S3: [3], S4: [4], S5: [5], S6: [6], S7: [7], S8: [8], S9: [9], S10: [10], S11: [11], S12: [12], S13: [13], S14: [14], S15: [23], S16: [19], S17: [21], S18: [22], S19: [24], S20: [25], S21: [26], S22: [27], S23: [28], S24: [29], S25: [30], S26: [31], S27: [32], S28: [33], S29: [34], S30: [35], S31: [36], S32: [37], S33: [38], S34: [39], S35: [40], S36: [41], S37: [42], S38: [43], S39: [44], S40: [45], S41: [46], S42: [47], S43: [60], S44: [48], S45: [15], S46: [49], S47: [50], S48: [51], S49: [52], S50: [53], S51: [54], S52: [17], S53: [16], S54: [55].

The comprehensive data extraction spreadsheet can be accessed at: https://bit.ly/SLM-LLM-Data-2026[3].

5 Findings

5.1 Corpus-Level Trends Before the RQ Analysis

Before answering the four research questions directly, three corpus-level observations are worth emphasizing.

First, the field is now primarily a systems literature rather than a pure model-scaling literature. Although a few studies still compare standalone small and large models directly, the dominant trend is to embed them inside a larger control strategy. Hybrid edge-cloud pipelines, retrieval-centered systems, expert routers, task decomposers, self-reflective workflows, and fusion modules are now common design choices. In other words, recent work increasingly treats model scale as one variable in a broader systems problem.

Second, hybridization is motivated by different constraints in different domains. In telecom and networking, the main concerns are deployment, context fidelity, and latency budgets. In healthcare and scientific tasks, the dominant concerns are reliability, interpretability, and domain specificity. In industrial monitoring, multimodal hybridization is driven by the need to combine coarse global understanding with fine-grained local detail. In optimisation and education, hybrid methods often serve as a way to recover reasoning quality from small models through feedback, routing, or targeted prompting.

Third, the evidence base is uneven. Accuracy is almost always reported; compute, latency, and cost are not. This matters because a large share of the literature frames hybridization as an efficiency solution, yet only a minority of studies expose sufficiently comparable numbers to test those claims rigorously. The result is a literature that often proves that hybridization *can* work, but less often proves how much it costs in practice under standardized conditions.

5.2 RQ1: Comparative Performance of Hybrid and Multi-Agent Systems

Across the corpus, hybrid and multi-agent architectures generally improve the performance ceiling of SLM-centered systems and, in several cases, also outperform larger monolithic baselines. The main reason is not that small models suddenly become universally strong, but that coordination mechanisms route the right subtasks, tokens, or contexts to the right component at the right moment.

The clearest evidence comes from edge-cloud collaboration. Hao et al. (S2) [2] show that token-level draft-verification can recover LLM-like quality on GSM8K while using only 25.8% of the

[3] Note for reviewers: This repository will be migrated to Zenodo for long-term archival during the camera-ready preparation phase.

full-cloud LLM cost. The study clarifies why this works: many tokens are easy enough for the on-device SLM, and only a small subset of difficult tokens must be verified or regenerated in the cloud. The key performance implication is that quality recovery does not require whole-query escalation. Token-level selectivity is enough.

This pattern also appears in wireless and mobile settings. Oh et al. (S45) [15] frame hybrid inference as a distributed speculative-decoding problem and show that uncertainty-aware skipping can preserve up to 97.54% of LLM accuracy. Their work is especially important for the present review because it ties performance directly to a measurable systems lever: if the SLM can estimate when its token is likely to be accepted, then the system can skip unnecessary communication and cloud inference without losing much quality.

Routing can do more than merely preserve LLM quality. In dialogue state tracking, OrchestraLLM (S6) [6] uses a retrieval-based router to select the most suitable expert among SLM and LLM pools. The study reports 82.46% turn-level joint goal accuracy on MultiWOZ and 68.09% on SGD, outperforming both the LLM and SLM baselines. The research makes the reason more explicit: SLMs and LLMs exhibit complementary strengths, with SLMs handling schema and formatting constraints well and LLMs contributing commonsense and broader contextual reasoning. This is an example of a hybrid system exceeding both ends of the scale spectrum because the system exploits complementarity rather than averaging behavior.

Domain-specific retrieval produces similarly strong results. TeleOracle (S4) [4] adapts Phi-2 to telecom question answering by combining fine-tuning, semantic chunking, hybrid retrieval, reranking, and context-window extension. In the reviewed corpus, this system reaches 81.20% accuracy, which is roughly 30% above the base Phi-2 model and competitive with much larger models. The study notes that telecom concepts can conflict with general-language usage. In such a setting, a small model supported by domain retrieval can outperform a larger but poorly grounded general-purpose model.

The reviewed literature also shows that hybridization is not limited to text-only inference. In EnerSafe-Cap (S5) [5], a lightweight video-language model and a multimodal large model are combined through dual-path sampling and prompt-guided fusion for industrial safety monitoring. This multimodal setting matters because it demonstrates that hybrid language-model design is also a coordination strategy over different visual abstractions. The lightweight path captures broad continuity, while the larger model focuses on key fine-grained moments. The performance gains over both SwinBERT and Qwen2-VL-2B suggest that hybridization can improve caption quality when different modalities or sampling strategies emphasize genuinely different information.

Several studies in the corpus further show that performance recovery can come from indirect support rather than direct LLM invocation at inference time. Mixed Distillation (S54) [55] uses both Chain-of-Thought and Program-of-Thought supervision to transfer reasoning competence into smaller models. In the evaluated findings, the resulting 7B models outperform GPT-3.5-Turbo on SVAMP. Likewise, FIPO (S52) [17] does not simply ask an LLM to solve the task directly; instead, it uses feedback-guided prompt optimisation to improve small-model problem formulation in mathematical optimisation. The result is a dramatic increase in Qwen-2.5 performance from 3% to 54% on LPWP. These examples show that hybridization can occur at the training-time or prompt-evolution level, not only at test-time inference.

Despite these positive results, the corpus does not support a simplistic conclusion that SLMs have "caught up" to LLMs in general. The strongest comparative results tend to appear in settings where at least one of the following is true: the task is domain-specific, the context can be retrieved and constrained, the output structure is well defined, the system can decompose the problem into

sub-parts, or escalation is possible when uncertainty is high. By contrast, open-ended reasoning, general medical question answering, and long-context understanding remain much harder for small models when those supporting conditions are absent.

5.3 RQ2: Efficiency, Cost, Latency, and Energy Trade-offs

The corpus strongly suggests that efficiency is one of the main motivations for SLM–LLM collaboration, but it also reveals that the evidence base is much weaker here than in pure performance evaluation. Out of 54 primary studies, 51 report task-quality measures, while only 17 report latency, 5 report cost, and 4 report energy. This asymmetry is itself a finding: hybrid systems are often justified as efficient, but only a minority of studies provide directly comparable measurements of that efficiency.

When cost is reported, the numbers are often compelling. Hao et al. (S2) [2] reduce cloud cost to 25.8%–31.2% of full LLM inference across three benchmark settings by escalating only a small subset of difficult tokens. Division-of-Thoughts (S53) [16] reports 83.57% lower API cost while maintaining competitive accuracy on reasoning and web-agent tasks. Token Level Routing (S46) [49] routes fewer than 7% of tokens to the cloud and claims more than 80% cost reduction relative to full LLM inference. These numbers point to a consistent design pattern: the largest savings come from selective escalation rather than from shrinking the entire pipeline.

Latency evidence is more complex. U-HLM (S45) [15] is one of the clearest studies because it reports both local and remote timing and links them to communication overhead. The SLM takes 24.6 ms and the LLM 104.6 ms in the reported setup, and uncertainty-aware skipping improves throughput by 2.54×. Division-of-Thoughts (S53) [16] improves average reasoning time by 66.12% through sub-task parallelism and selective allocation. In contrast, multimodal collaboration may improve accuracy at the cost of much slower per-instance processing: EnerSafe-Cap (S5) [5] reports approximately 19–29 seconds per video. These examples show that "hybrid" does not imply low latency by default. The latency effect depends on the granularity of cooperation and the burden of the coordination mechanism.

The reviewed studies also help explain why latency behaves differently across systems. In hybrid edge-cloud inference, latency overhead comes from token verification and communication. In U-HLM (S45) [15], the dominant bottleneck is vocabulary-distribution upload and repeated model evaluation. In OrchestraLLM (S6) [6], compute savings are reported in FLOPs rather than milliseconds because the router adds negligible cost relative to LLM execution. In TeleOracle (S4) [4], cross-encoder reranking is the latency bottleneck even though the core generator is small. The important systems lesson is that hybrid design moves cost around; it does not eliminate it.

Energy is the least mature evaluation axis in the corpus. Only four studies report energy-related information at all, and only two provide clearly numeric values. One study on energy-per-token (S34) [39] argues that a small model using Chain-of-Thought can be less energy-efficient than a larger model without such prompting. Another networking study (S16) [19] reports strong resource savings in near-real-time loops. These sparse reports are enough to show that energy cannot be inferred directly from parameter count, but they are not enough to support a high-confidence cross-study conclusion.

Overall, the efficiency evidence supports a cautious but clear synthesis. Hybridization is most beneficial when it suppresses unnecessary large-model calls, narrows the search space, or avoids sending every token, query, or subtask to the cloud. Efficiency gains are weakest or hardest to interpret when the coordination layer adds expensive retrieval, reranking, debate, or multimodal

processing that is not itself measured carefully. A mature literature on SLM–LLM systems will therefore need standardized reporting for latency, cloud usage, API cost, and energy, not just accuracy.

5.4 RQ3: Orchestration and Routing Mechanisms

The corpus contains a broad set of orchestration mechanisms, but they can be organized into a small number of recurring families. This taxonomy is important because the strongest studies in the review differ less by absolute model size than by how effectively they divide labor.

Token-level collaboration. Token-level orchestration appears in speculative or verification-based systems. Hybrid edge-cloud collaborative inference (S2) [2] uses the SLM as a drafter and the LLM as a verifier on hard tokens. U-HLM (S45) [15] adds an uncertainty-aware skipping mechanism so that even this token-level handshake is avoided whenever the SLM's token is likely to be accepted. Token-level routing is powerful because it preserves fine control over quality, but it can suffer from communication and synchronization overhead if too many tokens are escalated.

Query-level or example-level routing. OrchestraLLM (S6) [6] exemplifies retrieval-based expert routing at the dialogue level. Similar logic appears in customer-review analysis (S32) [37] and several question-answering systems. These designs are attractive when the system can identify "easy" and "hard" instances in a semantically meaningful way.

Retrieval-centered orchestration. RAG is a major control mechanism in the corpus even when the architecture is classified as hybrid rather than purely RAG. TeleOracle (S4) [4], RAP-RAG (S10) [10], ColBERT retrieval (S44) [48], and several domain QA systems use retrieval not only to expand context but also to constrain the action space of a smaller model. In these systems, retrieval functions as a coordination layer that substitutes for raw model scale by narrowing ambiguity and grounding the generated answer in domain evidence.

Planner-based decomposition and agentic workflows. Task decomposition is increasingly important in recent publications. Division-of-Thoughts (S53) [16] uses a task decomposer, dependency graph, and plug-and-play adapter to allocate sub-tasks between local and remote models. FIPO (S52) [17] uses an agentic workflow that generates structured feedback from multiple perspectives before updating the prompt of the small model. Flight-deck support (S11) [11], bioinformatics agents (S51) [54], and resilient task-execution frameworks all follow the same idea: if the task can be decomposed into specialists, small models can be retained in the loop even when a large model remains necessary somewhere in the pipeline.

Fusion and multimodal coordination. Not all orchestration is text-only. EnerSafe-Cap (S5) [5] combines coarse-grained and fine-grained visual semantics through prompt-guided fusion. Studies on molecular-property prediction (S3) [3] also fuse language-model embeddings with graph-based features. In these settings, orchestration is best understood as structured fusion between heterogeneous experts.

These patterns support a broader interpretation of orchestration. It is not merely an implementation detail added after choosing models. In the reviewed literature, orchestration is often the principal source of improvement. The best systems win because they know when to use the SLM,

when to ask for help, when to retrieve evidence, and when to decompose the task. This is why the same small models can perform poorly in monolithic form but competitively when embedded in a better control architecture.

5.5 RQ4: Domain Suitability of SLM-Centered Systems

The corpus shows that SLM competitiveness is domain-dependent rather than universal. Small models are most successful when at least one of the following conditions holds: the domain vocabulary is narrow, the task is structurally constrained, domain retrieval can ground the generation process, the output is partially symbolic or template-like, or the system can escalate uncertain cases to a larger model.

Telecom and networking. Telecom is one of the clearest success stories for SLM-centered systems. TeleOracle (S4) [4] and other reviewed telecom QA studies show that small models can perform well when standards documents are retrieved effectively and the generator is trained or prompted for domain fidelity. The reasons are intuitive: the domain is terminology-heavy, the relevant documents are accessible, and faithfulness matters at least as much as broad generative flexibility.

Healthcare and scientific domains. Healthcare results are mixed rather than uniformly positive. On the one hand, the review includes strong performance in radiology report labeling (S12) [12], chest X-ray reporting (S17) [21], VTE identification (S26) [31], multilingual medical summarization (S35) [40], and molecular property prediction (S3) [3]. These are tasks with domain structure, labeled outputs, or strong contextual priors. On the other hand, the corpus also contains medical question-answering (S21, S22) [26, 27] and diagnosis-generation (S19) [24] settings where larger models remain more capable or where reasoning prompts drastically inflate latency. The lesson is not that healthcare favors one model scale universally, but that smaller models work best when the output space is constrained and domain knowledge can be tightly anchored.

Reasoning, optimisation, and education. Reasoning-heavy tasks produce some of the most interesting hybrid results. Mixed Distillation (S54) [55] improves mathematical reasoning enough for 7B models to surpass GPT-3.5-Turbo on SVAMP. FIPO (S52) [17] shows that even very weak baseline SLM performance on optimisation formulation can be transformed through feedback-guided prompt evolution. Division-of-Thoughts (S53) [16] and question-answering agents demonstrate that decomposition and selective allocation can keep small models viable on reasoning workloads that would be implausible in a single-pass local-only setup. These studies suggest that small models are not inherently incapable of reasoning; rather, they are highly sensitive to scaffolding.

Code, software, and agentic tasks. The selected corpus includes several code-related or software-oriented studies, including code correction, type and call-graph analysis (S42) [47], offline robotic decision-making (S13) [13], and academic feedback generation (S23) [28]. These tasks often reward decomposition, specialist routing, or domain grounding more than sheer model size. Still, several reviewed systems note that performance drops as task complexity increases, especially when the agent must synthesize many steps or maintain long interaction histories.

Multimodal industrial monitoring. EnerSafe-Cap (S5) [5] illustrates a particularly important case because it is not a conventional text benchmark. In industrial video monitoring, the hybrid system outperforms both the lightweight and the larger baseline models because the task benefits from two complementary forms of attention: one to continuity and one to salient detail. This suggests that in multimodal domains, the question is often not whether the small model can replace the large one, but whether each can capture a different slice of the evidence.

Failure modes and limitations. The domain analysis also reveals several failure modes. Small models struggle most when the task requires broad open-domain knowledge, long-context synthesis without strong retrieval, or flexible reasoning over ambiguous instructions. They also become brittle when prompt structure matters too much or when coordination modules are themselves inaccurate. A central finding of the review is therefore that SLM suitability should be evaluated as a property of *system design plus domain*, not as a stable property of parameter count alone.

5.6 Cross-RQ Synthesis

Viewed together, the four research questions suggest that the literature is converging toward a specific systems principle: hybrid language-model design works best when the large model is treated as a selectively invoked capability rather than the default engine for every token, query, or subtask.

The first implication is architectural. Strong results are associated with explicit control layers such as routers, planners, retrievers, or fusion modules. This means that future progress is unlikely to come only from making SLMs slightly larger or LLMs slightly cheaper. It will also depend on better confidence estimation, task decomposition, retrieval selection, and cross-model scheduling.

The second implication is methodological. Because accuracy dominates reporting, many studies overstate practical efficiency. The corpus already contains cases where stronger orchestration clearly reduces cost or latency, but it also contains cases where coordination shifts the bottleneck rather than removing it. A mature evaluation protocol should therefore report resource metrics as first-class outcomes rather than optional extras.

The third implication is application-oriented. The most successful SLM-centered deployments are not generic substitutes for LLMs. They are domain-adapted systems in which the small model is grounded, specialized, or protected by escalation logic. This is why telecom, structured reasoning, radiology labeling, optimisation formulation, and industrial monitoring are so prominent in the positive findings of the corpus.

6 Threats to Validity

This review has several threats to validity that should be stated explicitly.

Internal validity. Although the review follows a PRISMA-guided workflow, study identification, screening, and extraction still involve researcher judgment. Borderline papers may have been interpreted differently under another reading of the inclusion criteria, especially in cases where the SLM component or the empirical contribution was not equally central throughout the paper. The representative PDF inspection used to deepen the narrative also adds a second judgment layer, because not every included study was read in the same depth. To reduce this risk, the manuscript keeps the synthesis tied to the predefined research questions and uses the same extraction matrix throughout.

Construct validity. Some of the core constructs in this review are not standardized across the literature. In particular, "SLM", "hybrid architecture", "multi-agent system", and "efficiency" are defined differently from paper to paper. A single hybrid label may refer to token-level verification, retrieval-grounded generation, planner-based decomposition, or multimodal fusion. Likewise, efficiency may mean latency, FLOPs, API cost, bandwidth reduction, or energy use. This threatens direct comparability and means that some studies grouped under the same high-level category differ substantially in mechanism.

Conclusion validity. The strongest limitation for cross-study inference is the uneven and inconsistent reporting of evaluation outcomes. Accuracy is reported in most papers, but latency, cost, and energy are available only in a minority, often with different units, hardware assumptions, or deployment settings. For that reason, the review can identify recurring patterns, but it cannot support a formal meta-analysis or strong quantitative claims about the size of efficiency gains across the whole corpus. This threat is reinforced by publication and reporting bias, since positive hybrid results are more likely to be published and summarized than negative or inconclusive findings.

External validity. The corpus is based on a Web of Science-centered search and a bounded review window. This improves transparency and reproducibility, but it may miss relevant studies from other databases, preprint platforms, or fast-moving venues. The field also evolves quickly, and the current corpus is heavily concentrated in 2025 and in a limited set of application domains. Therefore, the conclusions should be generalized cautiously beyond the covered time period, tasks, and deployment settings.

7 Conclusion

This paper presented a systematic literature review of 54 studies on SLMs, LLMs, and SLM–LLM collaboration across monolithic, hybrid, multi-agent, ensemble, and retrieval-centered architectures. The review was structured around four practical questions: comparative performance, efficiency, orchestration, and domain suitability. Across these four axes, the main pattern is consistent: SLMs are most competitive when they are supported by retrieval, routing, uncertainty estimation, task decomposition, prompt optimisation, or multimodal fusion rather than used as isolated replacements for larger models.

The findings also show that the literature is stronger in architectural innovation than in evaluation discipline. Many studies report accuracy improvements, but far fewer report latency, cost, or energy in a form that supports fair comparison. As a result, the current evidence is sufficient to show that hybridization can work, but often insufficient to show when it is the better deployment choice under realistic resource constraints. The review also suggests that the strongest positive results are concentrated in narrower or more structured settings such as telecom question answering, selected healthcare tasks, optimisation, and industrial monitoring, while open-ended reasoning and broad general-purpose use remain less convincing for SLM-centered systems.

For future research, three gaps stand out. First, the field needs more standardized reporting of resource metrics, hardware assumptions, and escalation rates. Second, it needs clearer operational definitions for SLM size ranges and architecture categories so that "hybrid" and "efficient" systems can be compared more reliably. Third, more domain-grounded evaluations are needed on failure cases, long-context behavior, orchestration overhead, and the conditions under which SLM-only, hybrid, and multi-agent designs stop scaling. In that sense, this SLR finds a literature that is already rich in system ideas, but still incomplete in measurement, comparability, and boundary-setting.

References

1. Tigani, S., et al.: CPU-Only Self Enhancing Authoring Copilot Design-Based Markov Decision Processes Orchestration and Qwen 3 Local LLM. Economies **13**(11), 520 (2025)
2. Hao, Z., Wei, J., Wang, T., Ren, J.: Hybrid SLM and LLM for Edge-Cloud Collaborative Inference. In: Proc. of the 2024 Workshop on Edge and Mobile Foundation Models (EdgeFM '24). ACM, New York, pp. 1–7 (2024)
3. Xie, L., Jin, Y., Xu, L., Chang, S., Xu, X.: Fusing Domain Knowledge with a Fine-Tuned Large Language Model for Enhanced Molecular Property Prediction. J. Chem. Theory Comput. **21**(14), 6743–6758 (2025). doi: 10.1021/acs.jctc.5c00605
4. Alabbasi, N., Erak, O., Alhussein, O., Lotfi, I., Muhaidat, S., Debbah, M.: TeleOracle: Fine-Tuned Retrieval-Augmented Generation with Long-Context Support for Networks. arXiv preprint arXiv:2411.02617 (2024)
5. Cao, Q., et al.: Toward Energy-Safe Industrial Monitoring: A Hybrid LLM Framework for Video Captioning. Appl. Sci. **15**(23), 12848 (2025)
6. Lee, C.H., Cheng, H., Ostendorf, M.: OrchestraLLM: Efficient Orchestration of Language Models for Dialogue State Tracking. In: Proc. of the 2024 Conf. of the North American Chapter of the Association for Computational Linguistics (NAACL-2024). ACL (2024)
7. Han, X., Sun, J., Yu, S., Qin, L., Luo, H., Li, Y.: CommentAgent: a LLM-powered agent framework for automated comment generation and opinion understanding. J. Supercomput. **81**(15), 1438 (2025). doi: 10.1007/s11227-025-07957-6
8. Aydın, S.K., Ali, R.H., Faiz, S., Khan, T.A.: An Integrated AI Framework for Personalized Nutrition Using Machine Learning and Natural Language Processing for Dietary Recommendations. Appl. Sci. **15**(17), 9283 (2025). doi: 10.3390/app15179283
9. Demir, E., Bilgin, M.: DPATransLLM: Detection of Pronominal Anaphora in Turkish Sentences Using Transformer-Based, Large Language Models and Hybrid Ensemble Approach. Appl. Sci. **15**(23), 12480 (2025). doi: 10.3390/app152312480
10. Ji, X., Chen, S., Shon, Y., et al.: RAP-RAG: A Retrieval-Augmented Generation Framework with Adaptive Retrieval Task Planning. Electronics **14**(1), 185 (2025)
11. Pacheco De Almeida Prado, J., de Oliveira, A.J.S., et al.: Enhancing Flight Deck Decision Support with Distributed GenAI: A Multi-Agent Approach. In: Proc. 2025 IEEE Aerospace Conf. (AERO), pp. 1–12. IEEE (2025)
12. Dorfner, F.J., Gadewar, M.R., et al.: Comparing Commercial and Open-Source Large Language Models for Labeling Chest Radiograph Reports. Radiology **313**(1), e240590 (2024)
13. Zhu, J., et al.: OFSLMs: Offline Fine-Tuned SLMs based on Hybrid Synthetic Knowledge for Robot Introspective Decision-making. IEEE ROBIO (2024)
14. Jeong, S., et al.: Test-Time Self-Adaptive Small Language Models for Question Answering. EMNLP (2023)
15. Oh, S., Kim, J., Park, J., Ko, S.W., Quek, T.Q.S., Kim, S.L.: Uncertainty-Aware Hybrid Inference with On-Device Small and Remote Large Language Models. In: Proc. 2025 IEEE Int. Conf. on Machine Learning for Communication and Networking (ICMLCN). IEEE (2025)
16. Shao, C., Hu, X., Lin, Y., Xu, F.: Division-of-Thoughts: Harnessing Hybrid Language Model Synergy for Efficient On-Device Agents. In: Proc. of the ACM Web Conference 2025 (WWW '25), pp. 1822–1833. ACM (2025)
17. Amarasinghe, P.T., Nguyen, S., Sun, Y., Alahakoon, D.: Feedback-integrated prompt optimiser for problem formulation. Sci. Rep. **15**, 43543 (2025)
18. Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., et al.: The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ **372**, n71 (2021)
19. Chatzistefanidis, G., et al.: Symbiotic Large and Small Language Models: A Survey. arXiv:2501.00000 (2025)

20. Chatzistefanidis, I., Nikaein, N.: Symbiotic Agents: A Novel Paradigm for Trustworthy AGI-driven Networks. Comput. Networks **273**, 111749 (2025)
21. Han, J., Kim, M., Kim, Y., Lee, W.H.: Self-Reflective Chest X-Ray Report Generation with Clinical-Aware Detection and Multilevel Readability. Journal of Medical Systems **49**, 190 (2025)
22. Pietrzak, Z., Mączka, K., Niemiec, M.: What if, Behind the Curtain, There Is Only an LLM? A Holistic Evaluation of TinyLlama-Generated Synthetic Cyber Threat Intelligence. Electronics **14**, 4971 (2025)
23. Weiss, Y., Buttery, P., Africa, D.D., Diehl Martinez, R.: Investigating ReLoRA: Effects on the Learning Dynamics of Small Language Models. arXiv preprint arXiv:2509.12960 (2025)
24. Gao, Y., Myers, S., Chen, S., Dligach, D., Miller, T., Bitterman, D.S., et al.: Uncertainty estimation in diagnosis generation from large language models: next-word probability is not pre-test probability. JAMIA Open **8**(1), ooae154 (2025)
25. Xiong, Y., Huang, M., Liang, X., Tao, M.: ResilioMate: A Resilient Multi-Agent Task Executing Framework for Enhancing Small Language Models. IEEE Access **13**, 86892–86911 (2025)
26. Ahmed, M., Arefeen, M.A., Zobaed, S.M., Rouf, M.T., Saha, S., Rahman, M.S.I.: Adapting Large Language Models for Arabic: Comparative Evaluation, Fine-Tuning, and Ethical Deployment. IEEE Access **13**, 12902–12928 (2025)
27. Rahman, M.S.I., Arefeen, M.A., Rouf, M.T., Ahmed, M., Zobaed, S.M., Saha, S., Samanta, S.: Prompt Design for Medical Question Answering with Large Language Models. IEEE Access **12**, 168239–168266 (2024)
28. Abbas, N., Atwell, E.: Cognitive Computing with Large Language Models for Student Assessment Feedback. Big Data Cogn. Comput. **9**(5), 112 (2025)
29. Han, X., Han, C., Chen, Y., Wei, J.: A Weighted Voting Approach for Traditional Chinese Medicine Formula Classification Using Large Language Models: Algorithm Development and Validation Study. JMIR Med. Inform. **13**, e60787 (2025)
30. Dodson, R.J.B., Dodson, C.R.S., Dodson, P.S., et al.: Explainable machine learning-based prediction of psoriatic arthritis flares using heterogenous real-world data for personalised patient care. Rheumatology **65**(2), 528–537 (2026)
31. Liu, X., Myers, S., Bitterman, D.S., Dligach, D., Miller, T., Afshar, M., Gao, Y.: Improving VTE Identification through Language Models from Radiology Reports: A Comparative Study of Mamba, Phi-3 Mini, and BERT. JAMIA Open **8**(1), ooae155 (2025)
32. Zeng, J., Liao, J., Sun, S., Zhang, G., Xu, C., Liu, H., Li, Y.: Multi-Agent Collaborative Reasoning via Cloud-Edge Framework. IEEE Network (2026). DOI: 10.1109/MNET.2026.3651036
33. Krishna, S.V., Saini, A.N.: Compos Mentis at SemEval-2024 Task 6: A Multi-Faceted Role-based Large Language Model Ensemble to Detect Hallucination. In: Proc. SemEval-2024, pp. 930–944. ACL (2024)
34. Chen, Y., Ma, S., Wang, S., Shi, J.: Integrating Reinforcement Learning and LLM with Self-Optimization Network System. IEEE Commun. Lett. **29**(1), 228–232 (2025)
35. Alabbasi, N., Erak, O., Alhussein, O., Lotfi, I., Muhaidat, S., Debbah, M.: Leveraging Fine-Tuned Retrieval-Augmented Generation with Long-Context Support: For 3GPP Standards. arXiv preprint arXiv:2411.02617 (2024)
36. Miller, K., Ghosh, S., Neumann, F.: Towards Optimizing a Retrieval Augmented Generation using Large Language Model on Academic Data. Bachelor Thesis, University of Adelaide (2024)
37. Alabbasi, N., Alhussein, O., Muhaidat, S., Debbah, M.: Do you actually need an LLM? Rethinking language models for customer reviews analysis. arXiv preprint arXiv:2411.02555 (2024)
38. Solat, F., Lee, J., Seif, M., Niyato, D., Poor, H.V.: Federated Learning-Enabled Hybrid Language Models for Communication-Efficient Token Transmission. IEEE Internet Things J. **12**(24), 53574–53587 (2025)
39. Wilhelm, P., Wittkopp, T., Kao, O.: Beyond Test-Time Compute Strategies: Advocating Energy-per-Token in LLM Inference. In: Proc. 5th Workshop on Machine Learning and Systems (EuroMLSys '25), pp. 1–7. ACM (2025)
40. Lee, C., Kumar, S., Vogt, K.A., Munshi, M., Tallapudi, P., Vogt, A., Awad, H., Khan, W.: Democratizing cost-effective, agentic artificial intelligence to multilingual medical summarization through knowledge distillation. Sci. Rep. **15**, 27619 (2025)

41. Jin, X., Wang, Z., Chen, J., Yang, L., Oh, B., Hwang, S., Li, J.: HLMEA: Unsupervised Entity Alignment Based on Hybrid Language Models. In: Proc. 39th AAAI Conf. on Artificial Intelligence (AAAI-25). AAAI (2025)
42. Huynh, T., Nguyen, D.: Collaborating Agents with Large Language Models for Question-Answering Tasks. In: Proc. 5th IEEE Int. Conf. on Software Engineering and Artificial Intelligence (SEAI), pp. 1–7. IEEE (2025)
43. Al Daoud, E., Al Daoud, L., Asassfeh, M., Al-Shaikh, A., Al-Sherideh, A.S., Afaneh, S.: Enhancing Cybersecurity with Transformers: Preventing Phishing Emails and Social Media Scams. In: Proc. IEEE Conf. on Dependable and Secure Computing (DSC), pp. 1–8. IEEE (2024)
44. Ying, S., Li, Z., Yu, M.: Beyond Words: Evaluating Large Language Models in Transportation Planning. arXiv preprint arXiv:2408.01633 (2024)
45. Pintelas, E., Koursaris, A., Livieris, I.E., Tampakas, V.: TextNeX: Text Network of eXperts for Robust Text Classification—Case Study on Machine-Generated-Text Detection. Mathematics **13**(10), 1555 (2025)
46. Esteban-Romero, S., Martín-Fernández, I., Gil-Martín, M., Fernández-Martínez, F.: Synthesizing Olfactory Understanding: Multimodal Language Models for Image-Text Smell Matching. Symmetry **17**(8), 1021 (2025)
47. Shivarpatna Venkatesh, A.P., Sunil, R., Sabu, S., Reis, M.S., Bodden, E., Amir, M.: An Empirical Study of Large Language Models for Type and Call Graph Analysis in Python and JavaScript. arXiv preprint arXiv:2410.00603 (2025)
48. Gichamba, A., Idris, T.K., Ebiyau, B., Nyberg, E., Mitamura, T.: ColBERT Retrieval and Ensemble Response Scoring for Language Model Question Answering. arXiv preprint arXiv:2408.10808 (2024)
49. She, J., Zheng, W., Liu, Z., Wang, H., Xing, E., Yao, H., Ho, Q.: Token Level Routing Inference System for Edge Devices. arXiv preprint arXiv:2504.07878 (2025)
50. Cielen, D., De Bock, K.W., Florès, L.: Exploring Zero-Shot SLM ensembles as an alternative to LLMs for sentiment analysis. Information Fusion **126**, 103666 (2026)
51. Bao, R., Xue, N., Sun, Y., Chen, Z.: Dynamic Quality-Latency Aware Routing for LLM Inference in Wireless Edge-Device Networks. arXiv preprint arXiv:2508.11291 (2025)
52. Chen, H., Deng, W., Yang, S., Xu, J., Jiang, Z., Ngai, E.C.H., Liu, J., Liu, X.: Toward Edge General Intelligence via Large Language Models: Opportunities and Challenges. arXiv preprint arXiv:2410.18125 (2025)
53. Li, M., Jiao, H., Zhou, T., Zhang, N., Peters, S., Lissitz, R.W.: Item Difficulty Modeling Using Fine-tuned Small and Large Language Models. In: Proc. 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2024), pp. 1–10 (2025)
54. Mehandru, N., Hall, A.K., Melnichenko, O., Dubinina, Y., Tsirulnikov, D., Bamman, D., Alaa, A., Saponas, S., Malladi, V.S.: BioAgents: Bridging the gap in bioinformatics analysis with multi-agent systems. Sci. Rep. **15**, 39036 (2025)
55. Li, C., Chen, Q., Li, L., Wang, C., Li, Y., Chen, Z., Zhang, Y.: Mixed Distillation Helps Smaller Language Model Better Reasoning. arXiv preprint arXiv:2312.10730 (2024)
56. Author(s): State of the Art and Future Directions of Small Language Models: A Systematic Review. Big Data & Cognitive Computing (2025)
57. Wang, F., et al.: A Comprehensive Survey of Small Language Models in the Era of Large Language Models: Techniques, Enhancements, Applications, Collaboration with LLMs, and Trustworthiness. arXiv:2411.03350 (2024)
58. Chen, Y., et al.: A Survey on Collaborative Mechanisms Between Large and Small Language Models. arXiv:2505.07460 (2025)
59. Wang, L., et al.: A Survey on Collaborating Small and Large Language Models for Performance, Cost-effectiveness, Cloud-edge Privacy, and Trustworthiness. arXiv:2510.13890 (2025)
60. Zhu, X., Li, J., Liu, Y., Ma, C., Wang, W.: Distilling Mathematical Reasoning Capabilities into Small Language Models. arXiv preprint arXiv:2401.11864 (2024)